

Unit 2: Data Link Layer and Medium Access Sub Layer

Topic 1: Error Detection and Error Correction Fundamentals

1. Introduction to Errors

When data is transmitted from one device to another, the received data may sometimes be different from the original data. This change is called an error.

Errors can occur because of noise, interference, signal problems, or other disturbances in the communication channel.

2. What Is an Error?

An error occurs when one or more bits in the received data are different from the corresponding bits sent by the sender.

For example, if the sender transmits 1011001 and the receiver gets 1010001, the received data contains an error because one bit has changed.

3. Types of Errors

The two basic types of errors are:

Type	Meaning
Single Bit Error	Only one bit in the data unit changes from its original value.
Burst Error	Two or more bits in a data unit are affected within a sequence of bits.

4. Single Bit Error

A single bit error occurs when only one bit changes from 0 to 1 or from 1 to 0.

Example:

Original Data	Received Data	Observation
1011010	1011110	One bit has changed.

Single bit errors are relatively simple to describe, but real communication systems can also experience errors affecting several nearby bits.

5. Burst Error

A burst error occurs when two or more bits in a data unit are affected within a sequence of bits.

Example:

Original Data	Received Data	Observation
1101011001	1100110001	Several bits are affected within the data sequence.

The affected bits do not necessarily have to be consecutive in value, but the error is described as a burst when the affected portion extends over a sequence of positions.

6. Causes of Errors

Errors may be caused by:

1. Electrical noise in the communication channel.
2. Electromagnetic interference.
3. Signal attenuation or distortion.
4. Problems in transmission equipment.
5. Interference from other signals.

6. Physical problems in the transmission medium.

7. Error Detection

Error detection is the process of determining whether the received data contains an error.

The sender adds extra information to the original data before transmission. The receiver uses this additional information to check whether the received data is correct.

If the receiver detects an error, it can take an appropriate action. Depending on the protocol, it may request retransmission or discard the damaged data.

8. Redundancy

Redundancy means adding extra bits to the original data for the purpose of detecting or correcting errors.

These extra bits do not represent new user information. They provide additional information that helps the receiver check the correctness of the received data.

The basic idea is simple: more carefully designed redundancy gives the receiver more information about whether the data has been changed during transmission.

9. Common Error Detection Techniques

Common techniques used for error detection include:

Technique	Basic Idea
Parity Check	Adds a parity bit to help detect certain bit errors.
Checksum	Uses a calculated value based on the data to help detect errors.
Cyclic Redundancy Check	Uses polynomial based calculation to detect errors in transmitted data.

These techniques use different methods, but their common purpose is to allow the receiver to determine whether the received data has been changed.

10. Error Correction

Error correction is the process of detecting an error and determining the correct value of the affected data so that the original information can be recovered.

For error correction, the transmitted data must contain sufficient additional information to allow the receiver to identify and correct certain errors.

11. Error Detection and Error Correction

Basis	Error Detection	Error Correction
Purpose	Determines whether an error has occurred.	Determines and corrects certain errors.
Additional Information	Uses redundancy to identify errors.	Uses sufficient redundancy to identify and correct errors.
Action After Detection	May request retransmission or take another recovery action.	Can correct the error directly when enough information is available.
Complexity	Generally simpler.	Generally requires more redundancy and processing.

12. How Error Detection Works

The basic process of error detection can be understood using the following steps:

1. The sender has the original data.
2. The sender calculates additional error detection information.
3. The additional information is attached to or associated with the data.
4. The complete data is transmitted through the communication channel.

5. The receiver receives the data and the additional information.
6. The receiver performs the required calculation or check.
7. If the result indicates a problem, the receiver concludes that an error has occurred.

13. How Error Correction Works

The basic process can be understood as follows:

1. The sender adds sufficient redundancy to the original data.
2. The data is transmitted through the communication channel.
3. Some bits may be changed during transmission.
4. The receiver examines the received data using the error correction method.
5. The receiver identifies the error when the method provides enough information.
6. The receiver corrects the affected data and recovers the original information.

14. Forward Error Correction

Forward Error Correction, or FEC, is a method in which the sender adds enough redundancy to the transmitted data so that the receiver can detect and correct certain errors without requesting retransmission.

FEC is useful when retransmission is difficult, expensive, or causes unacceptable delay.

15. Retransmission Based Error Control

Another approach is to detect an error and request the sender to transmit the data again.

This approach is used in error control protocols such as Automatic Repeat reQuest, or ARQ. The receiver detects an error and the sender retransmits the affected data according to the protocol rules.

16. Error Detection and Retransmission

In many communication systems, error detection and retransmission work together.

1. Sender transmits data with error detection information.
2. Receiver checks the received data.
3. If the data is correct, the receiver accepts it.
4. If an error is detected, the receiver informs the sender or does not acknowledge the data.
5. The sender retransmits the required data according to the protocol.

17. Difference Between FEC and ARQ

Feature	FEC	ARQ
Basic Approach	Corrects certain errors at the receiver using redundancy.	Detects errors and uses retransmission.
Retransmission	Normally not required for errors that can be corrected.	Required when damaged data must be transmitted again.
Delay	Can provide lower delay when correction is successful.	Retransmission can add delay.
Redundancy	Usually requires more redundancy for correction.	Uses redundancy mainly for error detection and control information.

18. Importance of Error Detection and Correction

1. Helps maintain data accuracy.
2. Reduces the effect of transmission errors.
3. Improves reliability of communication.
4. Allows damaged data to be detected.
5. Allows certain errors to be corrected.
6. Supports reliable data transfer protocols.

19. Simple Real Life Example

Suppose a teacher sends a student's marks through a communication system. If some bits change during transmission, the received marks may become incorrect.

An error detection technique can tell the receiver that the transmitted information has been changed. An error correction technique may be able to determine the correct data without asking for the message again.

If direct correction is not possible, a protocol can detect the error and request retransmission.

20. Important Definitions to Learn

Error: A condition in which the received data differs from the data transmitted by the sender.

Single Bit Error: An error in which only one bit changes from its original value.

Burst Error: An error in which two or more bits are affected within a sequence of transmitted bits.

Error Detection: The process of determining whether the received data contains an error.

Error Correction: The process of identifying and correcting certain errors in received data.

Redundancy: Extra information added to data to help detect or correct errors.

Forward Error Correction: A method in which redundancy allows the receiver to correct certain errors without retransmission.

ARQ: A method of error control in which detected errors can cause retransmission of data.

Topic 2: Block Coding and Hamming Distance

1. Introduction to Block Coding

Block coding is an error control technique in which a block of data bits is converted into a larger block by adding extra bits called redundant bits.

The extra bits provide additional information that helps the receiver detect or correct errors in the received data.

2. Basic Idea of Block Coding

In block coding, the original data is divided into fixed size blocks. Each block contains a fixed number of data bits.

For every block of data bits, some redundant bits are added. The resulting block is called a codeword.

The basic idea can be written as:

Data bits + Redundant bits = Codeword

If a data block contains k bits and the complete codeword contains n bits, then the number of redundant bits is n minus k .

3. Dataword and Codeword

The original block of data bits is called the dataword.

The block obtained after adding redundant bits is called the codeword.

Term	Meaning
Dataword	The original block of data bits.
Redundant Bits	Extra bits added for error control.
Codeword	The complete block containing data bits and redundant bits.

4. Example of Block Coding

Suppose the original dataword contains 4 bits. The sender adds 3 redundant bits. The resulting codeword contains 7 bits.

Item	Number of Bits
Dataword	4
Redundant bits	3

The receiver uses the rules of the coding scheme to check whether the received codeword is valid.

5. Why Are Redundant Bits Added?

Redundant bits are added because the receiver needs additional information to identify whether the received data has been changed during transmission.

1. They help detect errors.
2. They may help correct errors.
3. They improve the reliability of data communication.
4. They allow different codewords to be distinguished from one another.

6. Hamming Distance

Hamming distance is the number of bit positions in which two codewords are different.

To calculate the Hamming distance, compare the corresponding bits of the two codewords and count the positions where they are different.

7. Example of Hamming Distance

Consider the following two codewords:

Codeword 1: 101101

Codeword 2: 100001

Compare the bits at each position. The two codewords differ at two positions. Therefore, their Hamming distance is 2.

Hamming distance = 2

8. Minimum Hamming Distance

In a set of valid codewords, the minimum Hamming distance is the smallest Hamming distance between any two different valid codewords.

Minimum Hamming distance is important because it tells us how many errors a coding scheme can detect or correct.

9. Hamming Distance and Error Detection

If a coding scheme has a minimum Hamming distance of d , it can detect up to d minus 1 bit errors.

For example, if the minimum Hamming distance is 3, the coding scheme can detect up to 2 bit errors.

This happens because a received word must move through enough bit changes before it can become another valid codeword.

10. Hamming Distance and Error Correction

If a coding scheme has a minimum Hamming distance of d , it can correct up to floor of d minus 1 divided by 2 errors.

In simple form, the maximum number of correctable errors is calculated as:

Correctable errors = floor($(d - 1) / 2$)

For example, if the minimum Hamming distance is 5, the coding scheme can correct up to 2 bit errors.

11. Why Hamming Distance Is Important

1. It measures the difference between two codewords.
2. It helps determine the error detection capability of a code.
3. It helps determine the error correction capability of a code.
4. A larger minimum Hamming distance generally provides stronger error control.

12. Relationship Between Redundancy and Hamming Distance

Redundant bits provide additional information to the receiver. A suitable coding scheme uses these redundant bits to create valid codewords with useful separation between them.

This separation is measured using Hamming distance. Therefore, redundancy and Hamming distance are closely related to the error control capability of a coding scheme.

13. Error Detection Using Block Coding

The sender generates a valid codeword from the original dataword. When the receiver gets the codeword, it checks whether the received bit pattern is a valid codeword.

1. If the received pattern is a valid codeword, the receiver may accept it.
2. If the received pattern is not a valid codeword, an error is detected.
3. Depending on the coding method, the receiver may also determine the correct codeword.

14. Error Correction Using Block Coding

Some block codes contain enough redundancy to allow the receiver to identify the most likely original codeword. The receiver compares the received pattern with valid codewords and uses the coding rules to determine the correct data.

The ability to do this depends on the minimum Hamming distance of the code.

15. Simple Example of Detection and Correction

Suppose two valid codewords are separated by a Hamming distance of 3. If one bit changes during transmission, the received word may no longer be a valid codeword.

Because the minimum Hamming distance is 3, the code can detect up to 2 bit errors and can correct 1 bit error.

16. Difference Between Dataword and Codeword

Basis	Dataword	Codeword
Meaning	Original data block.	Data block after adding redundant bits.
Purpose	Contains the actual information.	Provides information for error control along with the actual data.
Size	Contains k bits.	Contains n bits, where n is greater than k.
Used During Transmission	Not normally transmitted alone in block coding.	Transmitted through the communication channel.

17. Difference Between Error Detection and Error Correction in Block Coding

Basis	Error Detection	Error Correction
Main Goal	Identify that the received data is incorrect.	Identify the error and recover the correct data.
Information Needed	Requires suitable redundancy.	Requires enough redundancy for correction.
Action	May lead to retransmission.	Can correct certain errors at the receiver.
Hamming Distance	d minimum allows detection of d minus 1 errors.	d minimum allows correction of floor of d minus 1 divided by 2 errors.

18. Simple Real Life Example

Suppose a teacher sends a four digit code to a student. Extra checking information is added to the code before sending it.

If one or more digits are changed during transmission, the checking information can help the student determine that the received code is not the original code.

If enough information was added, the student may also be able to determine the original code without asking the teacher to send it again.

19. Summary of the Main Ideas

1. Block coding divides data into fixed size blocks.
2. Extra redundant bits are added to each data block.
3. The original block is called the dataword.
4. The complete transmitted block is called the codeword.
5. Hamming distance counts the number of different bit positions between two codewords.
6. Minimum Hamming distance determines the error detection and correction capability of a code.
7. A minimum distance of d can detect up to d minus 1 errors.
8. A minimum distance of d can correct up to floor of d minus 1 divided by 2 errors.

Topic 3: Cyclic Redundancy Check

1. Introduction to CRC

Cyclic Redundancy Check, commonly called CRC, is an error detection technique used in data communication. CRC is used to determine whether the data received by the receiver has been changed during transmission. It is especially useful for detecting errors that may occur in a sequence of transmitted bits.

2. Basic Idea of CRC

In CRC, the sender performs a mathematical calculation on the data using a predefined generator. The result of this calculation is called the CRC value or remainder.

The sender adds this remainder to the original data and transmits the complete codeword.

The receiver performs a similar calculation. If the final remainder is zero, the received data is considered valid according to the CRC check. If the remainder is not zero, an error is detected.

3. Important Terms Used in CRC

Term	Meaning
Data	The original binary information that has to be transmitted.
Generator	A predefined binary value used for the CRC calculation.
Dividend	The binary data used in the division process.
Remainder	The result left after the required binary division.
CRC Bits	The remainder bits added to the original data before transmission.
Codeword	The complete transmitted block containing the data and CRC bits.

4. CRC and Polynomial Division

CRC uses binary division based on polynomial arithmetic. The calculation is performed using exclusive OR, or XOR, instead of ordinary subtraction.

In XOR operation, the result is 1 when the two input bits are different and 0 when they are the same.

A	B	A XOR B
0	0	0
0	1	1

1	0	1
1	1	0

5. Generator Polynomial

The generator polynomial represents the binary pattern used by the sender and receiver during the CRC calculation.

For example, a generator written as 1011 contains four bits. Its degree is 3 because the number of bits is one more than the degree.

The number of CRC bits added to the data is normally one less than the number of bits in the generator.

6. Steps Performed by the Sender

The sender follows a sequence of steps to calculate the CRC.

1. Take the original data.
2. Determine the number of CRC bits required from the generator.
3. Append the required number of zeros to the original data.
4. Divide the resulting binary sequence by the generator using XOR based division.
5. Obtain the remainder from the division.
6. Replace the appended zeros with the calculated remainder.
7. Transmit the resulting codeword.

7. Why Are Zeros Appended?

Zeros are appended to make space for the CRC remainder.

If the generator contains r plus 1 bits, r zeros are appended to the original data before division.

After the division, the r bit remainder replaces these zeros.

8. CRC Calculation Example

Consider the following simple example:

Data = 110101

Generator = 1011

The generator contains 4 bits, so 3 zeros are appended to the data.

Data after appending zeros = 110101000

The resulting sequence is divided by 1011 using XOR based binary division. The remainder obtained from this process becomes the CRC.

The exact remainder depends on carrying out the complete XOR division step by step.

9. XOR Based Binary Division

CRC division is different from normal decimal division. The main operation used is XOR.

Whenever the leftmost bit of the current portion is 1, the generator is XORed with that portion. When the leftmost bit is 0, the division moves forward without applying the generator at that position.

The process continues until all relevant bits have been processed.

10. Steps Performed by the Receiver

The receiver also performs a CRC calculation to check the received codeword.

1. Receive the complete codeword.
2. Use the same generator used by the sender.
3. Divide the received codeword by the generator using XOR based division.
4. Examine the final remainder.
5. If the remainder is zero, the data passes the CRC check.
6. If the remainder is not zero, an error is detected.

11. Error Detection Using CRC

CRC detects an error by checking whether the received codeword produces the expected remainder.

Receiver Result	Meaning
Remainder is zero	No error is detected by the CRC check.
Remainder is not zero	An error is detected.

A zero remainder means that the received codeword satisfies the CRC division rule. It does not mean that every possible physical transmission problem is mathematically impossible. CRC is an error detection technique, not a guarantee that every error will be detected.

12. Complete Working of CRC

1. Sender selects a generator.
2. Sender appends zeros to the data.
3. Sender performs XOR based division.
4. Sender obtains the CRC remainder.
5. Sender adds the remainder to the data and forms the codeword.
6. Sender transmits the codeword.
7. Receiver receives the codeword.
8. Receiver divides the codeword by the same generator.
9. Receiver checks the remainder.
10. A nonzero remainder indicates that an error has been detected.

13. Why Must the Same Generator Be Used?

The sender and receiver must use the same generator because the receiver needs to perform the same CRC rule used by the sender.

If different generators are used, the receiver may obtain an incorrect result even when the transmitted data is unchanged.

14. Advantages of CRC

1. CRC is effective for detecting many common transmission errors.
2. It is suitable for detecting burst errors.
3. The calculation can be implemented efficiently in digital hardware and software.
4. It provides stronger error detection than simple parity checking for many error patterns.
5. It is widely used in data communication and computer networks.

15. Limitations of CRC

1. CRC detects errors but does not normally correct them.
2. A detected error may require retransmission when a reliable protocol is being used.
3. The ability to detect errors depends on the selected generator.
4. CRC does not provide complete protection against every possible error pattern.

16. CRC Compared with Parity Check

Basis	Parity Check	CRC
Basic Method	Uses one or more parity bits.	Uses polynomial based XOR division.
Complexity	Simple.	More advanced.
Error Detection	Detects certain error patterns.	Can detect a wide range of error patterns.
Burst Errors	Limited capability.	Strong capability for many burst errors.

Use	Useful for simple error checking.	Widely used for stronger error detection.
-----	-----------------------------------	---

17. Simple Real Life Example

Suppose a teacher sends an important binary message to a student. Before sending it, the teacher calculates a CRC value and attaches it to the message.

During transmission, some bits may change. When the student receives the message, the student performs the same CRC calculation.

If the remainder is not zero, the student knows that the received message has been affected by an error.

18. Key Formula and Rules

If the generator contains r plus 1 bits:

Number of appended zeros = r

The sender calculates the remainder after XOR based division and uses that remainder as the CRC bits.

At the receiver, a nonzero remainder indicates that an error has been detected.

19. Simple CRC Flow

1. Original data
2. Append zeros
3. Divide by generator using XOR
4. Find remainder
5. Add remainder to data
6. Transmit codeword
7. Receiver divides by the same generator
8. Check the remainder
9. Detect error if the remainder is not zero

Topic 4: Flow Control and Error Control

1. Introduction

In data communication, the sender and receiver must work together properly so that data can be transferred correctly and efficiently.

Two important responsibilities of the data link layer are flow control and error control.

Flow control manages the rate at which data is sent. Error control deals with data that may be damaged or lost during transmission.

2. What Is Flow Control?

Flow control is a technique used to control the rate of data transmission between a sender and a receiver.

Its main purpose is to make sure that a fast sender does not send data faster than the receiver can process or store.

3. Why Is Flow Control Needed?

The sender and receiver may have different processing speeds and different amounts of available buffer space.

1. A sender may be able to transmit data very quickly.
2. A receiver may process the data more slowly.
3. The receiver may have limited buffer space.
4. Continuous transmission without control may cause the receiver buffer to become full.
5. Flow control helps prevent data from being lost because of receiver overload.

4. Simple Example of Flow Control

Suppose a teacher gives questions to a student much faster than the student can write the answers. After some time, the student cannot keep up.

Similarly, if a sender transmits frames faster than the receiver can process them, the receiver may become overloaded.

Flow control helps maintain a suitable transmission rate between the two devices.

5. What Is Error Control?

Error control is a set of techniques used to detect and handle errors that occur during data transmission.

The purpose is to make sure that the receiver obtains correct and complete data.

6. Why Do Errors Occur?

Errors may occur because of noise, interference, signal distortion, transmission problems, or other disturbances in the communication channel.

As a result, a transmitted frame may be changed, lost, or received incorrectly.

7. Main Functions of Error Control

1. Detect whether an error has occurred.
2. Identify that a frame has been lost or damaged when the protocol can determine this.
3. Recover the correct data.
4. Request retransmission when required.
5. Prevent damaged data from being accepted as correct data.

8. Relationship Between Flow Control and Error Control

Flow control and error control are different functions, but they often work together in reliable data communication.

Flow Control	Error Control
Controls the rate of transmission.	Controls the recovery of damaged or lost data.
Prevents receiver overload.	Helps maintain data correctness.
Deals mainly with sender and receiver speed.	Deals mainly with errors, loss, acknowledgement, timeout, and retransmission.
Uses techniques such as Stop and Wait and Sliding Window.	Uses techniques such as error detection and retransmission protocols.

9. Flow Control and Buffering

A buffer is a temporary storage area used to hold data while it waits to be processed.

When the sender is faster than the receiver, data can build up in the receiver buffer.

Flow control helps keep the amount of incoming data within the receiver's ability to store and process it.

10. Flow Control Using Stop and Wait

In Stop and Wait flow control, the sender sends one frame and waits for a response from the receiver before sending the next frame.

1. Sender sends one frame.
2. Receiver receives and processes the frame.
3. Receiver sends an acknowledgement.
4. Sender receives the acknowledgement.
5. Sender sends the next frame.

This method is simple because only one frame is allowed to be outstanding at a time.

11. Flow Control Using Sliding Window

In Sliding Window flow control, the sender can send multiple frames before waiting for individual acknowledgements, subject to the allowed window size.

The window represents the number of frames that the sender is permitted to send without waiting for further permission.

This makes better use of the communication channel than sending one frame and waiting after every frame.

12. Error Control Using Acknowledgement

An acknowledgement is a response sent by the receiver to indicate that a frame has been received or handled according to the protocol.

Acknowledgements are useful in error control because they provide feedback to the sender.

When an expected acknowledgement is not received, the sender may use a timeout and retransmit the required frame according to the protocol.

13. Timeout

A timeout is a waiting period after which the sender assumes that the expected response has not arrived.

If the acknowledgement does not arrive within the required time, the sender may retransmit the frame.

Timeout is therefore an important part of many error control protocols.

14. Retransmission

Retransmission means sending a frame again when the previous transmission is considered unsuccessful.

A frame may be retransmitted when it is damaged, lost, or when the sender does not receive the expected acknowledgement.

15. Error Control and ARQ

Automatic Repeat reQuest, commonly called ARQ, is an error control approach in which the receiver detects transmission problems and the sender retransmits data when required.

ARQ commonly uses acknowledgements, timeouts, and retransmission.

Stop and Wait ARQ, Go Back N ARQ, and Selective Repeat ARQ are important ARQ based protocols.

16. Stop and Wait ARQ

Stop and Wait ARQ combines the simple Stop and Wait method with error control.

1. Sender sends one frame.
2. Sender waits for an acknowledgement.
3. If the frame is received correctly, the receiver sends an acknowledgement.
4. If the frame is damaged or the acknowledgement does not arrive within the required time, the sender retransmits the frame.
5. The process continues for the remaining frames.

17. Sliding Window and Error Control

Sliding Window protocols can provide both flow control and error control.

The sender can have several frames in transmission, while acknowledgements and retransmission rules help manage errors.

Go Back N ARQ and Selective Repeat ARQ are examples of Sliding Window based error control protocols.

18. Difference Between Flow Control and Error Control

Basis	Flow Control	Error Control
Main Purpose	Controls the amount and rate of data sent to the receiver.	Detects and handles damaged or lost data.

Main Problem	Difference between sender speed and receiver capacity.	Errors, loss, and unsuccessful transmission.
Important Mechanisms	Window size, acknowledgements, and transmission rules.	Error detection, acknowledgements, timeout, and retransmission.
Examples	Stop and Wait and Sliding Window.	Stop and Wait ARQ, Go Back N ARQ, and Selective Repeat ARQ.

19. Difference Between Flow Control and Error Control in Simple Words

Flow control asks: How much data should the sender send so that the receiver can handle it?

Error control asks: What should we do if the data is damaged or lost during transmission?

Therefore, flow control is mainly concerned with the speed and amount of transmission, while error control is mainly concerned with the correctness and successful delivery of data.

20. Complete Example

Suppose a sender has many frames to transmit to a receiver. The sender is faster than the receiver, so sending every frame immediately could overload the receiver.

Flow control limits how many frames can be sent at a time.

Now suppose one of the transmitted frames is damaged by noise. Error control detects the problem using the protocol rules and may cause the damaged frame to be retransmitted.

In this way, flow control manages the amount of data being sent, while error control manages transmission errors.

21. Main Concepts to Remember

1. Flow control prevents the sender from overwhelming the receiver.
2. Error control helps provide reliable transmission.
3. Acknowledgement provides feedback from the receiver to the sender.
4. Timeout helps the sender detect that an expected response has not arrived.
5. Retransmission means sending data again.
6. Stop and Wait can be used for flow control and, with ARQ, for error control.
7. Sliding Window can provide efficient flow control and error control.
8. Go Back N ARQ and Selective Repeat ARQ are important Sliding Window error control protocols.

Topic 5: Stop and Wait Protocol

1. Introduction

Stop and Wait is a simple protocol used for controlling the flow of data between a sender and a receiver.

The basic rule is very simple: the sender sends one frame and then stops sending. It waits for a response from the receiver before sending the next frame.

2. Why Is Stop and Wait Needed?

A sender may be faster than a receiver. If the sender continuously sends frames, the receiver may not be able to process all of them.

Stop and Wait prevents this problem by allowing the sender to send only one frame at a time and wait for the receiver's response.

3. Basic Working of Stop and Wait

1. The sender prepares a frame.
2. The sender sends the frame to the receiver.
3. The sender stops sending further frames.

4. The receiver receives and processes the frame.
5. The receiver sends an acknowledgement.
6. The sender receives the acknowledgement.
7. The sender sends the next frame.

4. Acknowledgement

An acknowledgement is a response sent by the receiver to tell the sender that the frame has been received successfully.

The acknowledgement gives the sender permission to continue with the next frame.

Without this feedback, the sender would not know whether it should continue or wait.

5. Stop and Wait Example

Step	Sender	Receiver
1	Sends Frame 1	Waits for the frame
2	Waits	Receives Frame 1
3	Waits	Sends acknowledgement
4	Receives acknowledgement	Waits for the next frame
5	Sends Frame 2	Receives Frame 2

6. Stop and Wait as Flow Control

In basic Stop and Wait flow control, only one frame is allowed to be sent before the sender waits for an acknowledgement.

This prevents the sender from sending too much data too quickly.

The receiver gets enough time to process the current frame before the sender sends another frame.

7. What Happens If the Frame Is Lost?

If the frame is lost during transmission, the receiver does not receive it and therefore cannot send the expected acknowledgement.

The sender waits for the acknowledgement. If the acknowledgement does not arrive within the required time, the sender may assume that the transmission was unsuccessful and retransmit the frame when error control is being used.

8. Timeout

A timeout is a period for which the sender waits for the expected acknowledgement.

If the acknowledgement does not arrive before the timeout expires, the sender takes the required recovery action.

In Stop and Wait ARQ, this recovery action is normally retransmission of the frame.

9. Stop and Wait ARQ

Stop and Wait ARQ combines Stop and Wait flow control with error control.

ARQ stands for Automatic Repeat reQuest. It allows the sender to retransmit a frame when the receiver indicates a problem or when the expected acknowledgement is not received.

10. Working of Stop and Wait ARQ

1. The sender sends one frame.
2. The sender starts a timer.
3. The receiver receives the frame.
4. If the frame is correct, the receiver sends an acknowledgement.
5. The sender receives the acknowledgement before the timer expires.
6. The sender then sends the next frame.
7. If the frame is lost or damaged, the expected acknowledgement may not arrive.
8. When the timer expires, the sender retransmits the frame.

11. Frame Loss and Retransmission

Suppose Frame 1 is sent but is lost before reaching the receiver.

1. Sender sends Frame 1.
2. Frame 1 is lost.
3. Receiver does not receive Frame 1.
4. Receiver does not send the expected acknowledgement.
5. Sender waits until the timeout occurs.
6. Sender retransmits Frame 1.

12. What Happens If the Acknowledgement Is Lost?

Sometimes the receiver receives the frame correctly, but the acknowledgement is lost before reaching the sender.

1. Sender sends Frame 1.
2. Receiver receives Frame 1 correctly.
3. Receiver sends an acknowledgement.
4. The acknowledgement is lost.
5. Sender does not receive the acknowledgement.
6. The timer expires.
7. Sender retransmits Frame 1.
8. The receiver must recognize that the retransmitted frame is a duplicate and avoid delivering the same data twice.

13. Sequence Numbers

Sequence numbers are used to help the receiver distinguish a new frame from a retransmitted duplicate frame.

In a simple Stop and Wait ARQ system, a small sequence number field can be associated with each frame.

For example, sequence numbers 0 and 1 can be used alternately.

Frame	Sequence Number
First frame	0
Second frame	1
Third frame	0
Fourth frame	1

The receiver uses the sequence number to determine whether a received frame is new or is a duplicate of a frame already accepted.

14. Positive Acknowledgement

A positive acknowledgement indicates that the receiver has successfully received the expected frame.

After receiving the correct acknowledgement, the sender can move forward and transmit the next frame.

15. Duplicate Frames

A duplicate frame may occur when a frame reaches the receiver correctly but its acknowledgement is lost.

The sender retransmits because it did not receive the acknowledgement. The receiver uses sequence numbers to identify the duplicate and avoids delivering the same data again.

16. Advantages of Stop and Wait

1. It is simple to understand and implement.
2. It requires only a small amount of buffering.
3. It provides a simple method of flow control.
4. Stop and Wait ARQ can provide reliable transmission using acknowledgement, timeout, and retransmission.

17. Limitations of Stop and Wait

1. It can be slow when the communication path has a large delay.
2. The sender remains idle while waiting for an acknowledgement.
3. Only one frame can normally be outstanding at a time.
4. The available communication channel may not be used efficiently when the round trip delay is large.

18. Stop and Wait Compared with Sliding Window

Basis	Stop and Wait	Sliding Window
Frames in Transmission	Normally one frame at a time.	Multiple frames may be sent before waiting for all acknowledgements.
Efficiency	Lower when communication delay is large.	Generally more efficient for long delay paths.
Buffer Requirement	Small.	Usually larger.
Complexity	Simple.	More complex.
Flow Control	Simple.	Controlled using a window.
Error Control	Can use acknowledgement, timeout, and retransmission.	Can use acknowledgement and retransmission rules with a moving window.

19. Stop and Wait and Stop and Wait ARQ

Feature	Stop and Wait	Stop and Wait ARQ
Main Purpose	Flow control.	Flow control and error control.
Acknowledgement	Used to allow the next frame to be sent.	Used for flow control and to support error recovery.
Timeout	Not essential to the basic flow control idea.	Used to detect a missing expected acknowledgement.
Retransmission	Not part of the basic flow control mechanism.	Used when a frame must be sent again.

20. Simple Real Life Example

Imagine a teacher gives one question to a student and waits for the student to say that the question has been received before giving the next question.

The teacher is the sender and the student is the receiver. The student's response acts like an acknowledgement.

If the response does not arrive, the teacher waits for a suitable period and may repeat the question. This is similar to Stop and Wait ARQ.

21. Complete Working in Simple Words

Stop and Wait follows a simple rule: send one frame, wait for the acknowledgement, and then send the next frame. When error control is added, the sender also uses a timer. If the expected acknowledgement is not received, the sender retransmits the frame.

Sequence numbers help the receiver identify duplicate frames caused by retransmission.

Topic 6: Sliding Window Protocol

1. Introduction

Sliding Window is a flow control and error control technique that allows the sender to transmit more than one frame before waiting for acknowledgements.

Unlike Stop and Wait, the sender does not have to stop after every frame. A group of frames can be sent according to the size of the window.

2. Why Is Sliding Window Needed?

Stop and Wait is simple, but the sender may remain idle while waiting for an acknowledgement. This can reduce the use of the communication channel, especially when the delay between sender and receiver is large.

Sliding Window improves efficiency by allowing several frames to be sent before the sender has to wait for further acknowledgements.

3. Basic Idea of a Window

A window is a group of frame positions that the sender is currently allowed to use for transmission.

As acknowledgements are received, the window moves forward. This movement is why it is called a Sliding Window.

4. Sender Window

The sender window represents the frames that the sender is currently permitted to send.

For example, if the window size is 4, the sender may send up to four frames before it must wait for further acknowledgement information.

Window Size	Frames Allowed Initially
1	One frame
2	Two frames
4	Four frames
8	Eight frames

5. Receiver Window

The receiver window represents the frames that the receiver is prepared to accept according to the protocol being used.

The exact receiver window behavior depends on the particular Sliding Window protocol.

6. Working of Sliding Window

1. The sender has a sequence of frames to transmit.
2. The sender is allowed to transmit frames within the current window.
3. The receiver receives the frames and sends acknowledgements according to the protocol.
4. As acknowledgements are received, the sender moves the window forward.
5. New frames can then enter the available part of the window.
6. The process continues until all required data has been transmitted.

7. Example of Window Movement

Suppose the window size is 4 and the frames are numbered 0, 1, 2, 3, 4, 5, and so on.

Stage	Frames in Current Window
Initial	0, 1, 2, 3
After acknowledgement of frame 0	1, 2, 3, 4
After acknowledgement of frame 1	2, 3, 4, 5
After acknowledgement of frame 2	3, 4, 5, 6

The window moves forward as earlier frames are acknowledged. New frame positions become available for transmission.

8. Sequence Numbers

Sliding Window protocols use sequence numbers to identify frames.

A sequence number allows the sender and receiver to keep track of which frames have been sent, received, acknowledged, or are still expected.

The number of available sequence numbers and the permitted window size depend on the particular protocol.

9. Flow Control Using Sliding Window

Sliding Window provides flow control by limiting the number of frames that can be outstanding at the same time. If the receiver cannot accept more data, the sender must not continue sending beyond the permitted window. Therefore, the window acts as a limit on the amount of unacknowledged data that can be sent.

10. Acknowledgement

Acknowledgement tells the sender which transmitted data has been received according to the protocol. Acknowledgements allow the sender to move the window forward and make room for new frames.

11. Error Control Using Sliding Window

Sliding Window can also be used for error control. If a frame is damaged or lost, the protocol can use acknowledgement rules, timeout, and retransmission to recover the missing data.

Different Sliding Window protocols use different retransmission strategies.

Go Back N ARQ retransmits a group of frames beginning with the affected frame, while Selective Repeat ARQ retransmits only the frames that need retransmission.

12. Timeout in Sliding Window

A timer may be associated with transmitted frames. If the expected acknowledgement is not received within the required time, the sender can take the retransmission action specified by the protocol.

The exact timer and acknowledgement behavior depends on the Sliding Window protocol.

13. Window Size

Window size determines how many frames can be sent without waiting for further acknowledgement information. A larger window can allow more frames to be in transmission at the same time. This can improve channel utilization when the communication path has significant delay.

However, larger windows also require appropriate buffering and sequence number management.

14. Sliding Window Compared with Stop and Wait

Basis	Stop and Wait	Sliding Window
Frames Sent Before Waiting	Normally one.	Multiple frames can be sent according to the window size.
Efficiency	Lower when delay is high.	Generally higher for delayed communication paths.
Buffering	Usually small.	Usually requires more buffering.
Complexity	Simple.	More complex.
Flow Control	One frame at a time.	Uses a window to control outstanding frames.
Error Control	Can use retransmission in Stop and Wait ARQ.	Can use Go Back N ARQ or Selective Repeat ARQ.

15. Advantages of Sliding Window

1. It allows multiple frames to be transmitted before waiting.
2. It makes better use of the communication channel.
3. It can provide efficient flow control.
4. It can support error control and retransmission.
5. It is suitable for communication paths where waiting after every frame would waste time.

16. Limitations of Sliding Window

1. It is more complex than Stop and Wait.
2. It requires sequence number management.
3. It generally requires more buffering.

4. The window size must be selected and managed correctly.
5. Error recovery rules can become more complex when multiple frames are in transmission.

17. Simple Real Life Example

Imagine a teacher has to give ten questions to a student. With Stop and Wait, the teacher gives one question and waits for confirmation before giving the next one.

With Sliding Window, the teacher can give four questions first if the allowed window size is four. As the student confirms the questions, the teacher can give new questions.

This allows the teacher to continue working without waiting after every single question.

18. Complete Working in Simple Words

Sliding Window allows the sender to send several frames before waiting for all of them to be acknowledged.

The number of frames allowed is controlled by the window size. When acknowledgements arrive, the window moves forward and new frames can be sent.

Because several frames can be active at the same time, Sliding Window can use the communication channel more efficiently than Stop and Wait.

19. Main Concepts to Remember

1. Sliding Window allows multiple frames to be sent before waiting.
2. The window represents the currently allowed transmission positions.
3. The sender window controls the frames that can be sent.
4. The receiver window represents what the receiver can accept according to the protocol.
5. Acknowledgements allow the sender window to move forward.
6. Window size controls the amount of outstanding data.
7. Sliding Window can provide both flow control and error control.
8. Go Back N ARQ and Selective Repeat ARQ are important Sliding Window based protocols.

Topic 7: Go Back N ARQ

1. Introduction

Go Back N ARQ is an error control protocol based on the Sliding Window technique.

ARQ stands for Automatic Repeat reQuest. The main idea is that the sender can transmit several frames without waiting after every frame. If an error is detected, the sender retransmits the affected frame and the required following frames.

2. Why Is Go Back N ARQ Needed?

Stop and Wait ARQ sends one frame and waits for its acknowledgement. This can be inefficient when the communication path has a large delay.

Go Back N ARQ allows multiple frames to be transmitted before waiting. This improves channel utilization while still providing error control.

3. Basic Idea of Go Back N ARQ

In Go Back N ARQ, the sender is allowed to send a group of frames within its current window.

If a frame is lost or damaged, the receiver does not accept that frame as correct. The sender then goes back to the affected frame and retransmits that frame and the following frames that were sent after it.

This is the reason for the name Go Back N.

4. Window in Go Back N ARQ

Go Back N ARQ uses a sender window to control how many frames can be outstanding at one time.

For example, if the window size is 4, the sender may initially send four frames before it needs further acknowledgement information.

Window Size	Initial Frames
4	0, 1, 2, 3
5	0, 1, 2, 3, 4
8	0, 1, 2, 3, 4, 5, 6, 7

5. Sequence Numbers

Each frame is identified using a sequence number.

Sequence numbers allow the sender and receiver to identify the order of frames and determine which frames have been received or acknowledged.

The sequence number field has a finite range, so the protocol manages sequence numbers cyclically.

6. Cumulative Acknowledgement

Go Back N ARQ commonly uses cumulative acknowledgements.

A cumulative acknowledgement confirms the successful receipt of a sequence of frames up to a particular point. For example, if the receiver has correctly received frames 0, 1, 2, and 3 in order, an acknowledgement can indicate that these frames have been received and the sender may continue with later frames.

7. Normal Working of Go Back N ARQ

1. The sender sends several frames within the current window.
2. The receiver receives the frames in the expected order.
3. The receiver sends acknowledgements.
4. The sender receives the acknowledgements.
5. The sender moves its window forward.
6. New frames enter the window and can be transmitted.

8. What Happens When a Frame Is Lost?

Suppose the sender transmits frames 0, 1, 2, 3, and frame 2 is lost.

The receiver receives frame 0 and frame 1 correctly. When frame 3 arrives, the receiver is still expecting frame 2.

The receiver does not accept frame 3 as the next correct frame in the sequence. The sender eventually learns that frame 2 needs retransmission.

9. Retransmission in Go Back N ARQ

When an error or loss is detected, the sender retransmits the affected frame and all required frames that were transmitted after it within the affected range.

For example, if frame 2 is the missing frame and frames 3 and 4 were already sent, the sender may retransmit frame 2, frame 3, and frame 4.

Situation	Action
Frame 0, 1, 2 received correctly	Continue transmission.
Frame 2 lost or damaged	Recover starting from frame 2.
Frames 3 and 4 were sent after frame 2	They may also be retransmitted.

10. Timeout

A timer is used to detect when the expected acknowledgement has not arrived within the required time.

If the timer expires, the sender retransmits the required frames according to the Go Back N rules.

The timeout mechanism is important because a lost frame or lost acknowledgement may otherwise prevent the sender from moving forward.

11. Receiver Behavior

The receiver in Go Back N ARQ normally expects frames in sequence.

If an out of order frame arrives because an earlier frame is missing, the receiver generally does not accept it as the next valid frame.

The receiver continues to indicate the last correctly received sequence position according to the acknowledgement rules of the protocol.

12. Example of Go Back N ARQ

Frame	Sender Action	Receiver Result
0	Send	Received correctly
1	Send	Received correctly
2	Send	Lost
3	Send	Arrives while frame 2 is expected
2	Retransmit	Received correctly
3	Retransmit	Received correctly

The exact acknowledgement sequence depends on the implementation, but the important idea is that recovery begins from the missing or damaged frame.

13. Go Back N ARQ and Sliding Window

Go Back N ARQ is a Sliding Window based protocol.

Sliding Window Concept	Go Back N ARQ
Multiple frames can be outstanding.	Multiple frames can be transmitted before waiting.
Window controls transmission.	Sender window controls the number of outstanding frames.
Acknowledgements move the window.	Acknowledgements allow the sender window to advance.
Error recovery depends on protocol rules.	Frames from the affected point may be retransmitted.

14. Advantages of Go Back N ARQ

1. It is more efficient than Stop and Wait for many communication paths.
2. Multiple frames can be transmitted before waiting.
3. It provides flow control through the window.
4. It provides error control through acknowledgements and retransmission.
5. Its operation is simpler than protocols that individually retransmit every damaged frame.

15. Limitations of Go Back N ARQ

1. Correct frames may be retransmitted when an earlier frame is lost or damaged.
2. Retransmission can waste bandwidth.
3. The protocol is more complex than Stop and Wait.
4. It requires sequence number and window management.
5. Performance can decrease when errors occur frequently.

16. Go Back N ARQ Compared with Stop and Wait ARQ

Basis	Stop and Wait ARQ	Go Back N ARQ
Frames Sent Before Waiting	Normally one.	Multiple frames within the window.
Retransmission	Retransmits the affected frame.	Retransmits the affected frame and required following frames.

Efficiency	Lower for long delay paths.	Generally higher when errors are not frequent.
Complexity	Simple.	More complex.
Window	One frame is normally active.	Uses a Sliding Window.
Acknowledgement	Usually associated with the individual frame exchange.	Commonly uses cumulative acknowledgement.

17. Go Back N ARQ Compared with Selective Repeat ARQ

Basis	Go Back N ARQ	Selective Repeat ARQ
Error Recovery	Retransmits the affected frame and following required frames.	Retransmits only the frames that need retransmission.
Receiver Handling	Normally focuses on frames in sequence.	Can buffer suitable out of order frames.
Bandwidth Use During Errors	Can use more bandwidth because correct frames may be retransmitted.	Can use bandwidth more efficiently during errors.
Complexity	Simpler.	More complex.
Buffering	Less receiver buffering is generally needed.	More receiver buffering is generally needed.

18. Simple Real Life Example

Imagine a teacher sends a set of questions numbered 1, 2, 3, 4, and 5 to a student. The student receives questions 1 and 2, but question 3 is missing. Questions 4 and 5 arrive afterward.

The student cannot properly continue from question 3 because the expected question is missing. The teacher is informed through the communication process and sends question 3 again along with the later questions according to the protocol.

This is similar to the basic idea of Go Back N ARQ.

19. Complete Working in Simple Words

Go Back N ARQ allows the sender to send multiple frames. If a frame is lost or damaged, the sender goes back to that frame and retransmits it along with the required frames that followed it.

Acknowledgements and timeout help the sender determine when transmission should continue and when recovery is required.

20. Main Concepts to Remember

1. Go Back N ARQ is based on Sliding Window.
2. The sender can send multiple frames before waiting.
3. Sequence numbers identify frames.
4. Cumulative acknowledgements are commonly used.
5. If a frame is lost or damaged, recovery begins from that frame.
6. The sender retransmits the affected frame and the required following frames.
7. Timeout helps detect a missing expected acknowledgement.
8. Go Back N ARQ provides both flow control and error control.

Topic 8: Selective Repeat ARQ

1. Introduction

Selective Repeat ARQ is an error control protocol based on the Sliding Window technique.

ARQ stands for Automatic Repeat reQuest. Selective Repeat ARQ allows multiple frames to be transmitted before waiting for all acknowledgements.

When an error is detected, only the damaged or missing frame is retransmitted instead of retransmitting all the frames that followed it.

2. Why Is Selective Repeat ARQ Needed?

Go Back N ARQ may retransmit several frames even when some of those frames were received correctly. This can waste bandwidth.

Selective Repeat ARQ reduces this unnecessary retransmission by retransmitting only the frames that require recovery.

3. Basic Idea of Selective Repeat ARQ

The sender can transmit multiple frames within its current window.

The receiver can accept suitable frames and keep them in a buffer while waiting for a missing frame.

When a frame is lost or damaged, the sender retransmits only that frame.

4. Working of Selective Repeat ARQ

1. The sender sends multiple frames within the allowed window.
2. The receiver receives the frames.
3. Correct frames are accepted.
4. The receiver can acknowledge correctly received frames according to the protocol.
5. If one frame is missing or damaged, the receiver identifies the problem when possible.
6. The sender retransmits only the required frame.
7. The receiver stores suitable out of order frames until the missing frame is received.
8. After the required frame is received, the receiver can deliver the frames in the correct order.

5. Example of Selective Repeat ARQ

Frame	Transmission	Receiver Action
0	Sent	Received correctly
1	Sent	Received correctly
2	Sent	Lost
3	Sent	Received and buffered
2	Retransmitted	Received correctly
3	Already buffered	Can now be processed in order

In this example, frame 3 does not need to be retransmitted because the receiver already has it. Only frame 2 is retransmitted.

6. Receiver Buffering

Buffering is an important feature of Selective Repeat ARQ.

If a frame arrives before an earlier frame that is still missing, the receiver can temporarily store the out of order frame.

For example, if frame 2 is missing but frame 3 arrives correctly, the receiver can store frame 3 until frame 2 is received.

7. Selective Acknowledgement

Selective Repeat ARQ can use acknowledgements that identify individual frames that have been received correctly.

This allows the sender to determine exactly which frames need retransmission.

The acknowledgement behavior depends on the particular implementation of the protocol.

8. Timeout

A timer is used to determine whether an expected acknowledgement has not arrived within the required time.

If the timer for a frame expires, the sender retransmits that particular frame according to the protocol.

This is different from Go Back N ARQ, where recovery may cause several following frames to be retransmitted.

9. Sequence Numbers

Sequence numbers are used to identify frames and maintain their order.

They help the sender and receiver determine which frames have been transmitted, received, acknowledged, or are still missing.

Because sequence numbers have a finite range, the protocol must manage them carefully when the numbers are reused.

10. Sender Window

The sender window defines the range of frames that the sender can have outstanding.

As individual frames are acknowledged, the sender can move the window forward and transmit new frames.

The window allows several frames to be in transmission at the same time.

11. Receiver Window

The receiver window defines the range of sequence numbers that the receiver is prepared to accept according to the protocol.

A suitable receiver window allows the receiver to accept and buffer frames that arrive out of order.

12. Selective Repeat and Error Recovery

The main strength of Selective Repeat ARQ is selective retransmission.

If frame 2 is damaged while frames 0, 1, 3, and 4 are correct, the sender does not need to send frames 0, 1, 3, and 4 again. It retransmits frame 2.

This saves bandwidth and reduces unnecessary retransmission.

13. Advantages of Selective Repeat ARQ

1. Only damaged or missing frames are retransmitted.
2. It uses the communication channel efficiently when errors occur.
3. Correctly received out of order frames can be buffered.
4. It reduces unnecessary retransmission.
5. It can provide better performance than Go Back N ARQ when errors are frequent.

14. Limitations of Selective Repeat ARQ

1. It is more complex than Go Back N ARQ.
2. The receiver needs buffering for suitable out of order frames.
3. More sequence number and acknowledgement management is required.
4. The sender and receiver need more control information.
5. Correct implementation requires careful handling of duplicate and delayed frames.

15. Selective Repeat ARQ Compared with Go Back N ARQ

Basis	Go Back N ARQ	Selective Repeat ARQ
Retransmission	Retransmits the affected frame and required following frames.	Retransmits only the affected or missing frames.
Out of Order Frames	Normally not accepted for immediate delivery.	Can be accepted and buffered according to the protocol.
Bandwidth Use	Can waste bandwidth during retransmission.	Uses bandwidth more efficiently during errors.
Receiver Buffer	Generally simpler and smaller.	Requires more buffering.
Complexity	Lower.	Higher.

Performance During Errors	Can decrease when many unnecessary frames are retransmitted.	Generally better when errors are not rare.
---------------------------	--	--

16. Selective Repeat ARQ Compared with Stop and Wait ARQ

Basis	Stop and Wait ARQ	Selective Repeat ARQ
Frames in Transmission	Normally one.	Multiple frames within the window.
Retransmission	Retransmits the affected frame.	Retransmits only frames that require recovery.
Efficiency	Lower when communication delay is high.	Generally higher because multiple frames can be transmitted.
Complexity	Simple.	More complex.
Buffering	Limited requirement.	Receiver may need to buffer out of order frames.

17. Detailed Example

Suppose the sender transmits frames 0, 1, 2, 3, and 4. Frame 2 is damaged during transmission.

1. Frames 0 and 1 are received correctly.
2. Frame 2 is damaged.
3. Frame 3 arrives correctly.
4. Frame 4 arrives correctly.
5. The receiver stores frames 3 and 4 because frame 2 is still missing.
6. The receiver informs the sender that frame 2 requires retransmission.
7. The sender retransmits only frame 2.
8. The receiver receives frame 2 and can then process the stored frames in the correct order.

18. Simple Real Life Example

Imagine a teacher sends five questions to a student. The student receives questions 1, 2, 4, and 5 correctly, but question 3 is missing.

Instead of sending questions 3, 4, and 5 again, the teacher sends only question 3.

The student keeps questions 4 and 5 safely stored until question 3 arrives. This is similar to the basic idea of Selective Repeat ARQ.

19. Complete Working in Simple Words

Selective Repeat ARQ allows multiple frames to be sent at the same time. If one frame is lost or damaged, only that frame is retransmitted.

The receiver can temporarily store suitable out of order frames. After the missing frame arrives, the receiver can arrange and process the data in the correct order.

20. Main Concepts to Remember

1. Selective Repeat ARQ is based on Sliding Window.
2. Multiple frames can be transmitted before waiting.
3. Only the required damaged or missing frames are retransmitted.
4. The receiver can buffer suitable out of order frames.
5. Sequence numbers identify the frames.
6. Acknowledgements help the sender identify successfully received frames.
7. Timeout can cause retransmission of a particular frame.
8. Selective Repeat ARQ is more complex than Go Back N ARQ but can use bandwidth more efficiently during errors.

Topic 9: Piggybacking

1. Introduction

Piggybacking is a technique used in two way data communication to improve the efficiency of acknowledgements.

Instead of sending a separate acknowledgement frame immediately, a receiver can attach the acknowledgement information to a data frame that it is already sending in the opposite direction.

2. Basic Idea of Piggybacking

In two way communication, both devices may need to send data to each other.

Suppose Device A sends a data frame to Device B. Device B receives it and needs to acknowledge it. If Device B also has data to send to Device A, it can place the acknowledgement information inside its outgoing data frame. This combined data and acknowledgement transmission is called piggybacking.

3. Why Is Piggybacking Needed?

Sending a separate acknowledgement for every received frame can create additional control traffic.

When both devices are sending data, acknowledgement information can be included with the data that is already being transmitted.

1. It reduces the number of separate acknowledgement frames.
2. It reduces communication overhead.
3. It can improve channel utilization.
4. It can make two way communication more efficient.

4. Working of Piggybacking

1. Device A sends a data frame to Device B.
2. Device B receives the frame.
3. Device B has data that it wants to send to Device A.
4. Device B places the acknowledgement information for A's frame into its outgoing data frame.
5. Device B sends the combined frame to Device A.
6. Device A receives the data and the acknowledgement information together.

5. Simple Example

Device A	Device B
Sends Data Frame 1	Receives Data Frame 1
Waits	Has data to send to A
Receives Data Frame 2 with acknowledgement	Sends Data Frame 2 with acknowledgement for Frame 1

The acknowledgement for the first frame is carried along with a data frame travelling in the opposite direction.

6. Piggybacking in Two Way Communication

Piggybacking is useful when both devices communicate with each other.

If only one device is sending data and the other device has no data to send in the opposite direction, there may be no suitable outgoing data frame in which to place the acknowledgement.

7. Delayed Acknowledgement

Piggybacking may involve waiting for a short period to see whether the receiver has data to send in the opposite direction.

If suitable data is available, the acknowledgement can be attached to that data frame.

The waiting period should not be so long that it unnecessarily delays the acknowledgement.

8. Piggybacking and Acknowledgement

Separate Acknowledgement	Piggybacked Acknowledgement
Acknowledgement is sent in a separate frame.	Acknowledgement information is carried with a data frame.
Creates additional control traffic.	Reduces the need for a separate acknowledgement frame.
Useful when there is no outgoing data.	Useful when the receiver also has data to send.
Simple approach.	More efficient in suitable two way communication.

9. Advantages of Piggybacking

1. Reduces the number of separate acknowledgement frames.
2. Reduces overhead in two way communication.
3. Improves channel utilization.
4. Can improve overall communication efficiency.
5. Allows data and acknowledgement information to travel together.

10. Limitations of Piggybacking

1. It is mainly useful when both devices have data to send.
2. Waiting for outgoing data can delay the acknowledgement.
3. The protocol must use suitable timing rules.
4. If the acknowledgement is delayed for too long, the sender may experience a timeout.

11. Piggybacking and Sliding Window

Piggybacking is commonly associated with two way communication protocols that use acknowledgements and Sliding Window techniques.

As data frames travel in both directions, acknowledgement information can be included with outgoing data frames.

This reduces separate control traffic while allowing the sender to receive acknowledgement information.

12. Piggybacking and Flow Control

Piggybacking itself is not a separate flow control protocol. It is a technique for carrying acknowledgement information efficiently.

The acknowledgement information can still be used by the underlying protocol to manage the flow of frames.

13. Piggybacking and Error Control

Acknowledgements are important in error control because they provide information about successful reception. Piggybacking can carry these acknowledgements along with outgoing data. If an acknowledgement indicates successful reception, the sender can move forward according to the protocol.

14. When Piggybacking Is Useful

1. Both devices are actively exchanging data.
2. Acknowledgement information is required.
3. An outgoing data frame is available within a suitable time.
4. Combining the acknowledgement with data reduces unnecessary control traffic.

15. When a Separate Acknowledgement Is Better

If the receiver has no data to send in the opposite direction, waiting for an outgoing data frame may unnecessarily delay the acknowledgement.

In such a situation, the receiver can send a separate acknowledgement instead of waiting too long.

16. Piggybacking Compared with Separate Acknowledgement

Basis	Separate Acknowledgement	Piggybacking
Acknowledgement	Sent in a separate frame.	Attached to an outgoing data frame.
Control Traffic	Higher because of separate acknowledgement frames.	Lower when suitable outgoing data is available.
Efficiency	Can be lower in two way communication.	Can be higher in two way communication.
Delay	Acknowledgement can be sent immediately.	Acknowledgement may wait briefly for outgoing data.
Best Use	Useful when there is no outgoing data.	Useful when both devices are exchanging data.

17. Simple Real Life Example

Suppose two teachers are discussing students over a phone call. Teacher A gives some information to Teacher B. Teacher B wants to say that the information was received and also wants to give some information to Teacher A. Instead of sending a separate message only to say that the information was received, Teacher B can include the acknowledgement in the same message that contains the new information.

This is similar to piggybacking because acknowledgement information is carried along with useful data.

18. Complete Working in Simple Words

Piggybacking means attaching acknowledgement information to a data frame that is already travelling in the opposite direction.

It is useful when both devices are sending data. It reduces separate acknowledgement traffic and can improve communication efficiency.

However, the acknowledgement should not be delayed for too long just to wait for outgoing data.

19. Main Concepts to Remember

1. Piggybacking is used in two way communication.
2. It combines acknowledgement information with an outgoing data frame.
3. It reduces separate acknowledgement traffic.
4. It can improve channel utilization.
5. It is most useful when both devices have data to exchange.
6. Waiting too long for outgoing data can delay acknowledgement.
7. Piggybacking is a technique, not a separate flow control protocol.
8. It can work with protocols that use acknowledgements and Sliding Window techniques.

Topic 10: Random Access and Multiple Access

1. Introduction

In a computer network, several devices may need to use the same communication channel. A method is therefore required to decide how different devices can access the shared channel.

Random access and multiple access protocols are used to manage access to a shared communication medium.

2. What Is Multiple Access?

Multiple access means allowing multiple devices to share the same communication channel.

When many devices use one common channel, rules are required so that the devices can transmit data without causing unnecessary interference.

3. Why Are Multiple Access Protocols Needed?

If several devices transmit on a shared channel at the same time, their signals may interfere with one another. This can result in a collision, which may cause the transmitted frames to become unusable.

1. Many devices may want to transmit at the same time.
2. All devices share the same communication medium.
3. Simultaneous transmission can cause collisions.
4. A method is required to control channel access.
5. Multiple access protocols provide rules for using the shared channel.

4. What Is Random Access?

Random access is a method in which devices compete for access to a shared communication channel.

A device does not receive a fixed permanent turn. It attempts to transmit when it has data to send, according to the rules of the particular random access protocol.

Because multiple devices may attempt transmission at similar times, collisions can occur.

5. Basic Idea of Random Access

1. A device has data to transmit.
2. The device attempts to access the shared channel.
3. Another device may also attempt transmission.
4. If only one device transmits successfully, the frame can be received.
5. If multiple devices transmit together, a collision may occur.
6. The devices follow the protocol rules to recover from the collision.

6. What Is a Collision?

A collision occurs when two or more devices transmit on the same shared communication channel at the same time in a way that causes their signals to interfere.

The receiver may not be able to correctly interpret the combined signals.

The affected devices may need to wait and retransmit according to the access protocol.

7. Simple Collision Example

Situation	Result
Device A transmits alone	Transmission can be successful.
Device B transmits alone	Transmission can be successful.
Device A and Device B transmit together	Collision may occur.
Devices retransmit according to protocol rules	One or more later transmissions may succeed.

8. Main Types of Multiple Access Methods

Multiple access methods can be designed in different ways.

Method	Basic Idea
Random Access	Devices compete for access to the shared channel.
Controlled Access	Devices use an organized method to decide who can transmit.
Channelization	The available channel is divided among users or transmissions.

This topic focuses on random access methods included in the data link layer and medium access sub layer.

9. Random Access Protocols

Important random access protocols include Pure ALOHA, Slotted ALOHA, CSMA, CSMA CD, and CSMA CA. These protocols use different methods to reduce, detect, or avoid collisions.

Protocol	Basic Idea
Pure ALOHA	A device transmits when it has data. Collisions are handled through retransmission rules.
Slotted ALOHA	Transmission begins only at defined time slots.
CSMA	A device senses the channel before transmitting.
CSMA CD	A device senses the channel and detects collisions during transmission.
CSMA CA	A device attempts to avoid collisions before transmission.

10. Pure ALOHA

In Pure ALOHA, a device can transmit a frame whenever it has data to send.

The device does not wait for a fixed time slot before beginning transmission.

If another device transmits at a conflicting time, a collision may occur.

After a collision, the affected device waits for a suitable random period and then attempts retransmission according to the protocol.

11. Slotted ALOHA

Slotted ALOHA divides time into equal slots.

A device can begin transmission only at the beginning of a time slot.

This restriction reduces the period during which certain collisions can occur and improves channel utilization compared with Pure ALOHA.

12. CSMA

CSMA stands for Carrier Sense Multiple Access.

Before transmitting, a device listens to the shared channel to determine whether another transmission is already taking place.

If the channel is sensed as busy, the device follows the rules of the particular CSMA method. If the channel is sensed as idle, the device may transmit.

13. CSMA CD

CSMA CD stands for Carrier Sense Multiple Access with Collision Detection.

It allows a device to sense the channel before transmission and also detect a collision during transmission.

When a collision is detected, the device stops the affected transmission and follows the required recovery procedure.

14. CSMA CA

CSMA CA stands for Carrier Sense Multiple Access with Collision Avoidance.

It attempts to reduce the chance of collisions before transmission by using mechanisms such as waiting and random backoff.

CSMA CA is particularly associated with wireless communication, where detecting collisions during transmission is difficult.

15. Random Backoff

Random backoff means that a device waits for a randomly selected period before trying to transmit again.

Random waiting helps reduce the probability that the same devices will retransmit at exactly the same time and collide again.

16. Difference Between Random Access and Controlled Access

Basis	Random Access	Controlled Access
Channel Access	Devices compete for access.	Access is organized using a control mechanism.
Fixed Turn	No permanent fixed turn is normally assigned.	Transmission opportunities are organized.
Collision	Collisions can occur.	The control method is designed to reduce or avoid simultaneous transmission.
Examples	ALOHA and CSMA based methods.	Methods such as polling and token based access.

17. Random Access Compared with Channelization

Basis	Random Access	Channelization
Basic Idea	Devices compete for shared access.	The communication resource is divided among transmissions.
Collision	Can occur.	The division is designed to prevent users from interfering in the assigned portions.
Resource Use	Shared dynamically.	Assigned using a channel division method.
Examples	ALOHA and CSMA.	FDMA, TDMA, and CDMA.

18. Challenges in Multiple Access

1. Multiple devices may request the channel at the same time.
2. Collisions can reduce successful transmission.
3. Retransmission can increase delay.
4. Devices need rules for deciding when to transmit.
5. The protocol should use the shared channel efficiently.
6. The protocol should provide reasonable access to different devices.

19. Importance of Collision Handling

Collision handling is an important part of random access protocols.

A good protocol should provide a way for devices to recover after a collision and try again without causing continuous repeated collisions.

Different protocols use different techniques for this purpose. These techniques are studied in detail in Pure ALOHA, Slotted ALOHA, CSMA, CSMA CD, and CSMA CA.

20. Simple Real Life Example

Imagine several students want to speak during a group discussion. If everyone speaks at the same time, the teacher may not understand anyone.

The students therefore need rules for accessing the discussion. A student may wait, listen, speak, or try again after another student has finished.

This is similar to multiple devices sharing a common communication channel.

21. Complete Working in Simple Words

Multiple access allows many devices to share one communication channel. Random access allows devices to compete for access when they have data to send.

If two or more devices transmit at the same time, a collision may occur. The access protocol provides rules for handling the collision and trying again.

Pure ALOHA, Slotted ALOHA, CSMA, CSMA CD, and CSMA CA are important random access protocols.

22. Main Concepts to Remember

1. Multiple access allows multiple devices to share a communication channel.
2. Random access allows devices to compete for channel access.
3. A collision can occur when multiple devices transmit at the same time.
4. Pure ALOHA allows transmission whenever a device has data.
5. Slotted ALOHA allows transmission at the beginning of time slots.
6. CSMA senses the channel before transmission.
7. CSMA CD detects collisions during transmission.
8. CSMA CA attempts to avoid collisions before transmission.
9. Random backoff gives a device a waiting period before another transmission attempt.

Topic 11: Pure ALOHA

1. Introduction

Pure ALOHA is a random access protocol used when multiple devices share a common communication channel. It is one of the simplest methods of controlling access to a shared channel. A device can transmit whenever it has data to send.

2. Basic Idea of Pure ALOHA

Pure ALOHA does not divide time into fixed transmission slots. A device can begin transmission at any time. Because different devices may transmit at the same time, collisions can occur.

When a collision occurs, the affected frame is retransmitted after a randomly selected waiting period.

3. Working of Pure ALOHA

1. A device gets a frame that it wants to transmit.
2. The device immediately transmits the frame.
3. Another device may also transmit at the same time.
4. If transmissions overlap in a way that causes interference, a collision occurs.
5. The sender does not receive the expected successful transmission indication.
6. The sender waits for a random period.
7. The sender retransmits the frame.
8. The process continues until the frame is successfully transmitted or the protocol stops trying.

4. Collision in Pure ALOHA

A collision occurs when two or more devices transmit frames whose transmission periods overlap in a way that prevents correct reception.

Pure ALOHA has a relatively large possibility of collision because a device can begin transmission at any time.

5. Vulnerable Period

The vulnerable period is the time interval during which another transmission can interfere with a frame being transmitted.

For Pure ALOHA, if the frame transmission time is T , the vulnerable period is $2T$.

This large vulnerable period is one reason why Pure ALOHA has lower efficiency than Slotted ALOHA.

6. Retransmission After Collision

After a collision, the devices involved should not immediately retransmit at exactly the same time.

Instead, a device waits for a randomly selected period before making another transmission attempt.

Random waiting reduces the chance of repeated collisions.

7. Random Backoff

Random backoff means selecting a waiting period randomly before retransmitting a frame.

For example, if two devices experience a collision, they may select different waiting periods. The device with the shorter waiting period can attempt transmission first.

If another collision occurs, the protocol may require another waiting period before a new attempt.

8. Acknowledgement

An acknowledgement can be used to inform the sender that the receiver has successfully received the frame.

If the sender does not receive the expected acknowledgement within the required time, it may assume that the transmission was unsuccessful and retransmit the frame according to the protocol rules.

9. Timeout

A timeout is the waiting period used by the sender to determine whether the expected acknowledgement has arrived.

If the acknowledgement is not received before the timeout expires, the sender may retransmit the frame after applying the required random waiting procedure.

10. Throughput of Pure ALOHA

Throughput is the rate at which useful frames are successfully transmitted through the channel.

For Pure ALOHA, the theoretical maximum throughput is approximately 18.4 percent of the channel capacity.

The maximum occurs when the offered load reaches the appropriate operating point of the protocol.

The theoretical throughput of Pure ALOHA is represented by the formula:

$$S = G e^{\text{raised to the power of minus } 2G}$$

Here, S represents throughput and G represents the average number of transmission attempts during one frame transmission time.

11. Why Is Pure ALOHA Less Efficient?

1. Devices can transmit at any time.
2. The vulnerable period is large.
3. Collisions can occur frequently when many devices are active.
4. Collided frames must be retransmitted.
5. Retransmission adds additional traffic to the channel.

12. Pure ALOHA and Slotted ALOHA

Basis	Pure ALOHA	Slotted ALOHA
Transmission Time	A device can transmit at any time.	A device begins transmission at the beginning of a time slot.
Synchronization	No slot synchronization is required.	Devices must follow common time slots.
Vulnerable Period	2T.	T.
Maximum Theoretical Throughput	Approximately 18.4 percent.	Approximately 36.8 percent.
Collision Probability	Higher.	Lower than Pure ALOHA under comparable conditions.
Complexity	Simpler.	Requires time slot synchronization.

13. Advantages of Pure ALOHA

1. It is simple to understand.
2. Devices can transmit whenever they have data.
3. It does not require time slot synchronization.

4. It is easy to implement conceptually.
5. It provides a basic approach to shared channel access.

14. Limitations of Pure ALOHA

1. Collision probability can be high.
2. Channel utilization is low compared with Slotted ALOHA.
3. Retransmissions create additional traffic.
4. Performance decreases when many devices compete for the channel.
5. The large vulnerable period reduces efficiency.

15. Pure ALOHA at Low and High Traffic

When only a few devices have data to send, collisions are less frequent and Pure ALOHA can work reasonably well.

When many devices are trying to transmit, collisions become more frequent. More retransmissions then create additional traffic, which can further reduce successful transmission.

16. Simple Real Life Example

Imagine several students are sitting in a classroom and can speak whenever they want. If only one student speaks, the teacher can understand the message.

If two students start speaking at the same time, their voices overlap and the teacher may not understand either message.

The students can wait for different random periods and try again. This is similar to the basic idea of Pure ALOHA.

17. Complete Working in Simple Words

Pure ALOHA allows a device to transmit whenever it has a frame.

If another device transmits during an overlapping period, a collision may occur. The affected device waits for a random period and then retransmits.

Because transmission can begin at any time, the chance of collision is relatively high and the maximum theoretical throughput is approximately 18.4 percent.

18. Main Concepts to Remember

1. Pure ALOHA is a random access protocol.
2. A device can transmit at any time.
3. Collisions can occur when transmissions overlap.
4. A random waiting period is used before retransmission.
5. The vulnerable period is $2T$.
6. The maximum theoretical throughput is approximately 18.4 percent.
7. Pure ALOHA is simpler but less efficient than Slotted ALOHA.

Topic 12: Slotted ALOHA

1. Introduction

Slotted ALOHA is a random access protocol used when multiple devices share a common communication channel.

It is an improved version of Pure ALOHA. The main difference is that Slotted ALOHA divides time into equal slots and allows a device to begin transmission only at the beginning of a time slot.

2. Basic Idea of Slotted ALOHA

In Pure ALOHA, a device can start transmitting at any time. In Slotted ALOHA, transmission can start only at the beginning of a predefined time slot.

This restriction reduces the period during which transmissions can collide.

All devices need to follow the same slot timing for this method to work correctly.

3. Time Slots

Time is divided into equal intervals called slots.

Each slot is large enough to carry one complete frame.

Time	Slot
0 to T	Slot 1
T to 2T	Slot 2
2T to 3T	Slot 3
3T to 4T	Slot 4

A device that has a frame ready for transmission waits until the beginning of the next appropriate slot before transmitting.

4. Need for Time Synchronization

Slotted ALOHA requires devices to agree on the boundaries of the time slots.

This means that the devices need a common timing reference so that they know when a slot starts and ends.

Without suitable synchronization, devices may begin transmission at incorrect times and the advantage of slots can be reduced.

5. Working of Slotted ALOHA

1. Time is divided into equal slots.
2. A device gets a frame to transmit.
3. The device waits for the beginning of a suitable time slot.
4. The device transmits the frame during that slot.
5. If no other device transmits in the same slot, the frame can be received successfully.
6. If two or more devices transmit in the same slot, a collision occurs.
7. After a collision, the affected devices wait for a random number of slots.
8. The devices then try to retransmit their frames.

6. Collision in Slotted ALOHA

A collision occurs when two or more devices transmit during the same time slot.

Since transmissions begin only at slot boundaries, a frame is not exposed to collisions that begin at arbitrary points inside its transmission interval.

7. Vulnerable Period

The vulnerable period is the time during which another transmission can interfere with a frame.

If the frame transmission time is T , the vulnerable period in Slotted ALOHA is T .

This is half the vulnerable period of Pure ALOHA, which is $2T$.

8. Retransmission After Collision

When a collision occurs, the affected devices do not immediately retransmit in the same way.

Each device waits for a randomly selected number of time slots before making another transmission attempt.

Random waiting helps reduce the probability that the same devices will collide again.

9. Random Backoff in Slotted ALOHA

Random backoff means selecting a waiting period before retransmitting.

The waiting period is measured in complete time slots.

For example, after a collision, one device may wait for two slots while another device may wait for five slots. The device with the shorter waiting period gets an opportunity to transmit earlier.

10. Acknowledgement

An acknowledgement can be used to indicate that a frame has been received successfully.

If the sender does not receive the expected acknowledgement within the required time, it can treat the transmission as unsuccessful and follow the retransmission procedure.

11. Timeout

A timeout is the period during which the sender waits for the expected acknowledgement.

If the acknowledgement is not received within the required time, the sender may assume that the frame was lost or collided and prepare for retransmission.

12. Throughput of Slotted ALOHA

Throughput represents the amount of useful data successfully transmitted through the shared channel.

The theoretical throughput of Slotted ALOHA is represented by:

$$S = G e^{-G}$$

Here, S represents throughput and G represents the average number of transmission attempts per slot.

The maximum theoretical throughput of Slotted ALOHA is approximately 36.8 percent.

This is approximately twice the maximum theoretical throughput of Pure ALOHA, which is approximately 18.4 percent.

13. Why Is Slotted ALOHA More Efficient?

1. Devices can transmit only at the beginning of a slot.
2. The vulnerable period is reduced from $2T$ to T .
3. A frame is not affected by a transmission that starts halfway through its transmission period.
4. The probability of certain overlapping collisions is reduced.
5. The maximum theoretical throughput is higher than Pure ALOHA.

14. Pure ALOHA and Slotted ALOHA

Basis	Pure ALOHA	Slotted ALOHA
Transmission Start	Can begin at any time.	Can begin only at the beginning of a slot.
Time Slots	Not required.	Required.
Synchronization	Not required.	Required.
Vulnerable Period	$2T$.	T .
Maximum Theoretical Throughput	Approximately 18.4 percent.	Approximately 36.8 percent.
Collision Possibility	Higher.	Lower under comparable traffic conditions.
Complexity	Simpler.	Requires time synchronization.

15. Example of Slotted ALOHA

Slot	Device A	Device B	Result
1	Transmits	No transmission	Successful transmission by A
2	No transmission	Transmits	Successful transmission by B
3	Transmits	Transmits	Collision

4	Waits	Waits	Backoff after collision
5	Transmits	Waits	A can transmit successfully

16. What Happens During a Collision?

1. Two or more devices transmit in the same slot.
2. The receiver cannot correctly decode the overlapping transmissions.
3. The transmissions are considered unsuccessful.
4. The affected devices select random waiting periods.
5. The devices attempt retransmission in later slots.

17. Advantages of Slotted ALOHA

1. It is more efficient than Pure ALOHA.
2. The vulnerable period is reduced.
3. The maximum theoretical throughput is approximately 36.8 percent.
4. The basic working is still relatively simple.
5. Random access is maintained while improving channel utilization.

18. Limitations of Slotted ALOHA

1. It requires time synchronization among devices.
2. Collisions can still occur.
3. Retransmissions create additional traffic.
4. Performance decreases when many devices compete heavily for the channel.
5. The maximum theoretical throughput is still below complete channel utilization.

19. Slotted ALOHA at Low and High Traffic

At low traffic, collisions are less frequent because fewer devices are trying to transmit.

As traffic increases, more devices may select the same slot. Collisions then become more frequent, causing more retransmissions.

Therefore, Slotted ALOHA performs best near its theoretical operating point and becomes less efficient under heavy offered load.

20. Simple Real Life Example

Imagine a classroom where students are allowed to speak only at the beginning of fixed time intervals. If one student speaks during a particular interval, the teacher can listen to that student.

If two students begin speaking at the same time interval, their voices may overlap and a collision occurs.

The students can wait for different numbers of intervals before trying again. This is similar to Slotted ALOHA.

21. Complete Working in Simple Words

Slotted ALOHA divides time into equal slots. A device can start transmission only at the beginning of a slot.

If only one device transmits in a slot, the transmission can be successful. If two or more devices transmit in the same slot, a collision occurs.

After a collision, the devices wait for random numbers of slots and then try again.

Because transmission starts only at slot boundaries, Slotted ALOHA has a smaller vulnerable period and a higher maximum theoretical throughput than Pure ALOHA.

22. Main Concepts to Remember

1. Slotted ALOHA is a random access protocol.
2. Time is divided into equal slots.

3. A device can begin transmission only at the beginning of a slot.
4. Time synchronization is required.
5. A collision occurs when multiple devices transmit in the same slot.
6. The vulnerable period is T .
7. The maximum theoretical throughput is approximately 36.8 percent.
8. Slotted ALOHA is more efficient than Pure ALOHA.

Topic 13: CSMA

1. Introduction

CSMA stands for Carrier Sense Multiple Access. It is a random access protocol used when many devices share the same communication channel.

The main idea of CSMA is simple. Before transmitting, a device first listens to the channel. It checks whether another device is already transmitting.

2. Meaning of Carrier Sensing

Carrier sensing means listening to the communication channel before sending data.

If the channel is busy, the device waits according to the rules of the particular CSMA method.

If the channel is idle, the device can attempt to transmit.

3. Why CSMA Is Needed

In a shared network, many devices may want to transmit at the same time. If all devices transmit without checking the channel, collisions can occur frequently.

CSMA tries to reduce collisions by making devices listen before they transmit.

4. Basic Working of CSMA

1. A device has a frame ready for transmission.
2. The device senses the communication channel.
3. If the channel is idle, the device follows the rules of its CSMA method and transmits.
4. If the channel is busy, the device waits according to the selected CSMA method.
5. If another device also begins transmission at a nearly similar time, a collision can still occur because of propagation delay.
6. The device then follows the required collision handling and retransmission procedure.

5. Channel States in CSMA

Channel State	Meaning	Basic Action
Idle	No device is currently detected transmitting.	The device can attempt transmission.
Busy	Another device is transmitting.	The device waits according to the CSMA method.

6. Why Collision Can Still Occur in CSMA

CSMA reduces collisions, but it cannot completely eliminate them.

Consider two devices that are located at different positions in a network. Both devices may sense the channel as idle at almost the same time because a transmission from one device has not yet reached the other device.

Both devices may then begin transmitting. Their signals can overlap and cause a collision.

This happens because signals need a certain amount of time to travel through the network. This time is called propagation delay.

7. Types of CSMA

The main types of CSMA covered here are:

1. 1 Persistent CSMA
2. Non Persistent CSMA
3. P Persistent CSMA

8. 1 Persistent CSMA

In 1 Persistent CSMA, the device continuously senses the channel.

If the channel is idle, the device transmits immediately.

If the channel is busy, the device keeps sensing the channel. As soon as the channel becomes idle, the device transmits.

The word persistent means that the device keeps trying to transmit as soon as the channel becomes available.

9. Working of 1 Persistent CSMA

1. The device senses the channel.
2. If the channel is idle, it transmits immediately.
3. If the channel is busy, it continues sensing.
4. When the channel becomes idle, it transmits immediately.
5. If several devices are waiting for the same busy channel, more than one device may transmit when the channel becomes idle, which can cause a collision.

10. Advantages of 1 Persistent CSMA

1. It is simple to understand and implement.
2. A device does not transmit while the channel is already detected as busy.
3. It can provide quick transmission when the channel becomes idle.

11. Limitations of 1 Persistent CSMA

1. Several waiting devices may transmit at the same time when the channel becomes idle.
2. This can increase the probability of collision.
3. Continuous sensing can increase the use of channel sensing resources.

12. Non Persistent CSMA

In Non Persistent CSMA, a device senses the channel before transmitting.

If the channel is idle, the device transmits.

If the channel is busy, the device does not continuously sense the channel. Instead, it waits for a randomly selected period and then senses the channel again.

This random waiting reduces the chance that many waiting devices will begin transmission at exactly the same time.

13. Working of Non Persistent CSMA

1. The device senses the channel.
2. If the channel is idle, it transmits.
3. If the channel is busy, it waits for a random period.
4. After waiting, it senses the channel again.
5. If the channel is idle, it transmits.
6. If the channel is still busy, it repeats the waiting and sensing process.

14. Advantages of Non Persistent CSMA

1. It can reduce collisions among devices waiting for a busy channel.

2. Random waiting spreads transmission attempts over time.
3. It avoids continuous sensing while the channel is busy.

15. Limitations of Non Persistent CSMA

1. A device may have to wait even after the channel becomes free.
2. This can increase delay.
3. Channel utilization may be lower in some situations.

16. P Persistent CSMA

P Persistent CSMA is mainly used with a slotted channel.

When the channel is found to be idle, the device transmits with probability P. With probability 1 minus P, it waits for the next slot and repeats the decision.

This method provides a balance between transmitting immediately and waiting.

17. Working of P Persistent CSMA

1. The device senses the channel.
2. If the channel is busy, the device waits.
3. If the channel is idle, the device generates a probability based decision.
4. With probability P, the device transmits.
5. With probability 1 minus P, the device waits for the next slot.
6. The process continues until the device transmits or the protocol requires another action.

18. Meaning of P

P is the probability of transmission when the channel is found to be idle.

For example, if P is 0.5, the device has a 50 percent probability of transmitting when the channel is idle and a 50 percent probability of waiting for the next slot.

19. Comparison of CSMA Methods

Feature	1 Persistent CSMA	Non Persistent CSMA	P Persistent CSMA
When channel is idle	Transmit immediately.	Transmit.	Transmit with probability P.
When channel is busy	Keep sensing.	Wait for a random period.	Wait until the channel is available according to the slot based procedure.
Main idea	Transmit as soon as channel becomes idle.	Wait randomly before sensing again.	Use a probability to decide whether to transmit.
Collision tendency	Can be higher when many devices are waiting.	Can be reduced by random waiting.	Can be controlled through the value of P.
Delay	Can be low when channel becomes free.	Can be higher because of random waiting.	Depends on P and network traffic.

20. CSMA and ALOHA

ALOHA allows a device to transmit without first sensing the channel. CSMA first checks the channel before transmission.

Basis	ALOHA	CSMA
Channel sensing	Not performed before transmission.	Performed before transmission.

Collision control	Uses retransmission after collisions.	Tries to reduce collisions by sensing the channel.
Basic idea	Transmit when a frame is ready.	Listen before transmitting.

21. CSMA and CSMA CD

CSMA reduces the probability of collision by sensing the channel before transmission.

CSMA CD adds collision detection. It checks the channel during transmission to detect a collision and then takes action such as stopping transmission and applying a backoff procedure.

Therefore, CSMA is the basic listen before transmission idea, while CSMA CD adds collision detection and collision handling.

22. Simple Example

Imagine three students sharing one microphone. Before speaking, each student listens to check whether someone else is already speaking.

If the microphone is free, a student can speak. If someone is speaking, the other students wait.

However, if two students are far apart and both think the microphone is free at nearly the same time, both may start speaking. Their voices can overlap. This is similar to a collision in CSMA.

23. Real Life Example

Consider a group of people sharing a single communication channel. Before speaking, each person listens to see whether someone else is speaking.

If nobody is speaking, a person can start. If someone is speaking, the person waits. If several people start at nearly the same time because of communication delay, their messages may interfere.

This gives a simple idea of how CSMA works.

24. Complete Working in Simple Words

CSMA follows the basic rule of listening before transmitting.

A device first checks the shared channel. If the channel is idle, it can transmit according to the rules of the selected CSMA method. If the channel is busy, the device waits.

Different CSMA methods decide differently what the device should do after finding a busy channel. In 1 Persistent CSMA, it keeps sensing and transmits when the channel becomes idle. In Non Persistent CSMA, it waits for a random period before sensing again. In P Persistent CSMA, it uses a probability to decide whether to transmit when the channel is idle.

CSMA reduces collisions, but collisions can still occur because of propagation delay.

25. Main Concepts to Remember

1. CSMA stands for Carrier Sense Multiple Access.
2. A device senses the channel before transmitting.
3. CSMA reduces the probability of collisions but does not completely remove them.
4. Propagation delay can cause collisions even when CSMA is used.
5. 1 Persistent CSMA transmits immediately when the channel becomes idle.
6. Non Persistent CSMA waits for a random period when the channel is busy.
7. P Persistent CSMA uses probability P to decide whether to transmit.
8. CSMA is the basic idea used before understanding CSMA CD and CSMA CA.

Topic 14: CSMA CD

1. Introduction

CSMA CD stands for Carrier Sense Multiple Access with Collision Detection. It is an extension of CSMA that not only checks the channel before transmission but also detects a collision while transmission is taking place. The main purpose of CSMA CD is to reduce the time wasted when a collision occurs and to provide a method for handling that collision.

2. Basic Idea of CSMA CD

CSMA CD follows the basic rule of CSMA: listen before transmitting.

The important additional feature is collision detection. A device continues monitoring the communication channel while it is transmitting.

If the device detects that another transmission has interfered with its transmission, it knows that a collision has occurred.

3. Need for CSMA CD

In a shared communication channel, two devices may sometimes begin transmitting at nearly the same time. Their signals can overlap and create a collision.

If a device continued transmitting the complete frame after a collision, channel time would be wasted.

CSMA CD allows the devices to detect the collision and stop the unsuccessful transmission early.

4. Working of CSMA CD

1. A device has a frame ready for transmission.
2. The device senses the channel.
3. If the channel is busy, the device waits.
4. If the channel is idle, the device starts transmitting.
5. While transmitting, the device continues monitoring the channel.
6. If no collision is detected, the frame transmission finishes successfully.
7. If a collision is detected, the device stops transmitting.
8. The device sends a jam signal so that other devices become aware of the collision.
9. The device waits for a backoff period.
10. After the backoff period, the device senses the channel again and attempts retransmission.

5. Carrier Sensing

Carrier sensing is the first important step in CSMA CD.

Before transmitting, a device listens to the shared channel.

If another device is already transmitting, the channel is considered busy and the device waits.

If the channel is idle, the device can begin transmission.

6. Collision Detection

Collision detection means checking the communication channel during transmission to determine whether another signal has interfered with the current transmission.

If the transmitted signal and the observed signal do not match as expected, the device can identify that a collision has occurred.

Once a collision is detected, the device stops transmitting the current frame.

7. Why Does a Collision Still Occur?

CSMA CD cannot completely prevent collisions because of propagation delay.

Suppose two devices are located far apart. Both devices may sense the channel as idle because the transmission from one device has not yet reached the other device.

Both devices may then begin transmitting. Their signals can meet somewhere in the shared network and cause a collision.

8. Jam Signal

After detecting a collision, the device sends a jam signal.

The purpose of the jam signal is to make the collision condition clear to the other devices sharing the channel.

After the jam signal, the device stops the current transmission and prepares for retransmission.

9. Backoff

After a collision, a device does not immediately retransmit. It waits for a certain period called the backoff period. The waiting period is selected according to the backoff procedure. A random waiting period helps reduce the chance that the same devices will retransmit together and cause another collision.

After waiting, the device senses the channel again and attempts transmission when permitted.

10. Random Backoff

Random backoff means that a device selects a waiting time from an allowed range instead of always waiting for exactly the same amount of time.

For example, after a collision, two devices may choose different waiting periods. The device with the shorter waiting period gets an earlier opportunity to transmit.

This reduces repeated collisions between the same devices.

11. Acknowledgement

An acknowledgement may be used by the communication system to confirm successful delivery of a frame.

The exact acknowledgement mechanism depends on the network technology and protocol being used. CSMA CD itself is mainly concerned with sensing the channel, detecting collisions, sending the jam signal, and applying backoff.

12. Step by Step Example

Step	Device A	Device B	Channel
1	Has a frame.	Has a frame.	Idle
2	Senses channel.	Senses channel.	Idle
3	Starts transmitting.	Starts transmitting almost simultaneously.	Collision occurs
4	Detects collision.	Detects collision.	Collision detected
5	Sends jam signal and stops.	Sends jam signal and stops.	Transmission stopped
6	Chooses backoff time.	Chooses backoff time.	Devices wait
7	Attempts again after its backoff.	Waits if its backoff is longer.	Transmission can proceed

13. CSMA CD Compared with Basic CSMA

Feature	CSMA	CSMA CD
Channel sensing	Yes.	Yes.
Collision detection during transmission	Not part of basic CSMA.	Yes.
Collision handling	Depends on the protocol.	Detect collision, send jam signal, apply backoff, and retransmit.

Main purpose	Reduce the chance of collision.	Reduce collision waste and handle detected collisions efficiently.
--------------	---------------------------------	--

14. CSMA CD Compared with CSMA CA

Basis	CSMA CD	CSMA CA
Full form	Carrier Sense Multiple Access with Collision Detection.	Carrier Sense Multiple Access with Collision Avoidance.
Main approach	Detect a collision after transmission begins.	Try to avoid collisions before transmission.
Typical use	Traditionally associated with shared wired Ethernet.	Commonly associated with wireless LANs.
After a collision or failure	Uses collision detection and backoff procedures.	Uses avoidance methods, waiting, and acknowledgement mechanisms.
Important idea	Detect the collision.	Try to avoid the collision.

15. Advantages of CSMA CD

1. It checks the channel before transmission.
2. It can detect collisions while transmission is taking place.
3. It stops an unsuccessful transmission after a collision is detected.
4. It uses backoff to reduce repeated collisions.
5. It improves the use of a shared channel compared with a method that does not sense the channel.

16. Limitations of CSMA CD

1. It cannot completely eliminate collisions.
2. Propagation delay can still cause collisions.
3. Collision detection is difficult in wireless environments because a device may not reliably detect another signal while transmitting.
4. Performance decreases when many devices compete for the same channel.
5. The protocol requires additional mechanisms for collision handling.

17. CSMA CD and Ethernet

CSMA CD was an important access method for traditional shared Ethernet networks.

In a traditional shared Ethernet network, multiple devices could use the same physical communication medium. CSMA CD helped coordinate access to that shared medium.

Modern switched full duplex Ethernet normally does not require CSMA CD because each link can provide a separate communication path and simultaneous transmission and reception can occur without the same type of shared medium collision.

18. Simple Real Life Example

Imagine several students sharing one microphone. Before speaking, each student listens to check whether someone else is speaking.

If the microphone appears free, a student starts speaking while continuing to listen.

If two students begin speaking almost together and their voices overlap, they realize that a collision has occurred.

Both stop, wait for different amounts of time, and then try again.

This gives a simple idea of CSMA CD.

19. Complete Working in Simple Words

CSMA CD first checks whether the communication channel is free. If the channel is busy, the device waits. If the channel is free, the device begins transmitting.

While transmitting, the device continues checking the channel. If the transmission completes without a collision, the process is successful.

If a collision occurs, the device detects it, stops the transmission, sends a jam signal, and waits for a backoff period. It then tries to transmit the frame again.

The main idea can therefore be remembered as: sense the channel, transmit, detect collision, stop, back off, and retransmit.

20. Main Concepts to Remember

1. CSMA CD stands for Carrier Sense Multiple Access with Collision Detection.
2. The device senses the channel before transmission.
3. The device also monitors the channel while transmitting.
4. A collision can still occur because of propagation delay.
5. When a collision is detected, the device stops transmitting.
6. A jam signal is sent after collision detection.
7. The device waits for a backoff period before retransmission.
8. Random backoff helps reduce repeated collisions.
9. CSMA CD is mainly associated with traditional shared wired Ethernet.

Topic 15: CSMA CA

1. Introduction

CSMA CA stands for Carrier Sense Multiple Access with Collision Avoidance. It is a random access method used mainly in wireless networks.

The main idea is to listen to the channel before transmission and then take additional steps to reduce the chance of a collision.

2. Why CSMA CA Is Needed

In a wireless network, collision detection is difficult. A wireless device normally cannot reliably transmit and listen to the same channel at the same time in the way required for collision detection.

Therefore, instead of detecting a collision after it happens, CSMA CA tries to avoid collisions before transmission. This is the main difference between collision detection and collision avoidance.

3. Basic Idea of CSMA CA

A wireless device first senses the channel.

If the channel is busy, the device waits.

If the channel is idle, the device does not always transmit immediately. It may wait for a required period and use a random backoff before transmitting.

The purpose is to reduce the probability that several devices will start transmitting at the same time.

4. Basic Working of CSMA CA

1. A device has a frame ready for transmission.
2. The device senses the wireless channel.
3. If the channel is busy, the device waits.
4. If the channel is idle, the device waits for the required idle period.
5. The device uses a random backoff procedure when required.
6. When the backoff reaches the required point, the device transmits.
7. The receiver receives the frame and can send an acknowledgement.

8. If the acknowledgement is not received, the sender assumes that the transmission was unsuccessful and follows the retransmission procedure.

5. Collision Avoidance

Collision avoidance means trying to reduce the possibility of collision before a device transmits.

CSMA CA uses waiting, channel sensing, random backoff, and acknowledgement mechanisms to reduce the chance of simultaneous transmissions.

Even with these methods, collisions cannot always be completely eliminated.

6. Interframe Space

After the channel has been sensed as idle, a wireless device may need to wait for a specified period before transmission.

This waiting period is called an interframe space.

It helps provide an organized method for controlling access to the wireless channel.

7. Random Backoff

Random backoff is an important part of CSMA CA.

When a device has to wait before transmission, it selects a random waiting value from an allowed range.

The waiting value is counted while the channel remains suitable for the backoff procedure.

If another device starts transmitting, the backoff process can pause. When the channel becomes available again, the device can continue its remaining backoff.

Random backoff reduces the chance that many devices will transmit at exactly the same time.

8. Example of Random Backoff

Suppose three wireless devices are waiting to transmit. Each device selects a different random backoff value.

Device	Selected Backoff	Basic Result
Device A	2 slots	Gets an earlier opportunity.
Device B	5 slots	Waits longer.
Device C	8 slots	Waits longer than A and B.

Device A reaches the transmission opportunity first if the channel remains available. The other devices continue waiting. This reduces the chance that all three devices will begin transmission together.

9. Acknowledgement

An acknowledgement, commonly called ACK, is a message sent by the receiver to indicate successful reception of a frame.

The sender waits for the expected acknowledgement after transmitting.

If the acknowledgement is received, the sender knows that the transmission was successful.

If the acknowledgement is not received within the required time, the sender can treat the transmission as unsuccessful and prepare for retransmission.

10. Timeout

A timeout is the period during which the sender waits for an expected acknowledgement.

If the acknowledgement does not arrive before the timeout expires, the sender can assume that the frame was lost or that a collision or other problem may have occurred.

The sender then follows the required retransmission procedure, usually involving another backoff.

11. RTS and CTS

CSMA CA can use two control messages called RTS and CTS.

RTS stands for Request to Send.

CTS stands for Clear to Send.

These messages can help reduce collisions in situations where wireless devices cannot hear each other directly.

12. Working of RTS and CTS

1. A sender wants to transmit a data frame.
2. The sender sends an RTS message.
3. The receiver responds with a CTS message if it is ready to receive.
4. Other nearby devices that hear the CTS can remain silent for the required transmission period.
5. The sender transmits the data frame.
6. The receiver sends an acknowledgement after successful reception.

13. Hidden Terminal Problem

The hidden terminal problem occurs when two wireless devices cannot hear each other but both can communicate with the same receiver.

For example, Device A and Device C may both be outside the sensing range of each other. Both may think the channel is free and transmit toward Device B at the same time.

Their transmissions can interfere at Device B.

RTS and CTS can help reduce this problem by informing other devices that the channel is going to be used.

14. Why CSMA CA Is Common in Wireless Networks

Wireless devices face conditions that make collision detection difficult.

A device may not be able to detect another transmission reliably while it is transmitting.

Wireless devices can also experience hidden terminal and exposed terminal situations.

Therefore, wireless communication uses collision avoidance techniques instead of relying on collision detection in the same way as traditional shared Ethernet.

15. CSMA CA and CSMA CD

Basis	CSMA CA	CSMA CD
Full form	Carrier Sense Multiple Access with Collision Avoidance.	Carrier Sense Multiple Access with Collision Detection.
Main idea	Try to avoid collisions before transmission.	Detect collisions after transmission begins.
Common association	Wireless LANs.	Traditional shared wired Ethernet.
Random backoff	Used to reduce simultaneous transmissions and after unsuccessful attempts.	Used after collision detection.
Acknowledgement	Commonly used to confirm successful wireless reception.	Not the defining feature of CSMA CD.
RTS and CTS	Can be used.	Not used as the normal collision handling method.

16. CSMA CA and ALOHA

Basis	ALOHA	CSMA CA
Channel sensing	Does not use carrier sensing before transmission.	Senses the channel before transmission.
Collision approach	Allows transmission and handles unsuccessful attempts through retransmission.	Attempts to avoid collisions before transmission.
Random waiting	Used after collisions.	Used as part of collision avoidance and retransmission procedures.
Typical use	Random access environments.	Wireless LAN environments.

17. Advantages of CSMA CA

1. It is suitable for wireless communication.
2. It tries to reduce collisions before transmission.
3. Random backoff reduces simultaneous transmission attempts.
4. Acknowledgements provide confirmation of successful reception.
5. RTS and CTS can help with the hidden terminal problem.

18. Limitations of CSMA CA

1. Collisions cannot be completely eliminated.
2. Waiting and backoff increase transmission delay.
3. RTS and CTS create additional control traffic when they are used.
4. Performance can decrease when many devices compete for the same wireless channel.
5. Wireless conditions can change because of interference, distance, and other factors.

19. Simple Example

Imagine several students sharing one classroom microphone. Before speaking, each student listens to check whether someone else is already speaking.

If the microphone is free, a student waits for the required short period and then chooses a random waiting count.

The student whose waiting count finishes first gets the opportunity to speak. The others wait.

This reduces the chance that several students will begin speaking at exactly the same time.

20. Real Life Example

Consider several mobile or wireless devices connected through a common wireless access point. Many devices may have data ready at the same time.

Each device senses the wireless channel and waits when necessary. A device may select a random backoff period before transmission. After sending its frame, it can wait for an acknowledgement.

This provides a simple way to understand how CSMA CA manages access to a shared wireless channel.

21. Complete Working in Simple Words

CSMA CA means listening before transmitting and trying to avoid collisions.

A wireless device first checks the channel. If the channel is busy, it waits. If the channel is idle, the device follows the required waiting procedure and uses random backoff when required.

After the backoff period, the device transmits. The receiver can send an acknowledgement to confirm successful reception.

If the acknowledgement is not received, the sender assumes that the transmission was unsuccessful and tries again after following the required waiting and backoff procedure.

RTS and CTS can also be used to reduce the hidden terminal problem.

The main idea is: sense the channel, wait, use random backoff, transmit, receive acknowledgement, and retransmit when necessary.

22. Main Concepts to Remember

1. CSMA CA stands for Carrier Sense Multiple Access with Collision Avoidance.
2. It is mainly associated with wireless LANs.
3. The device senses the channel before transmission.
4. It tries to avoid collisions instead of detecting them after transmission begins.
5. Random backoff is used to reduce simultaneous transmission attempts.
6. An acknowledgement can confirm successful reception.
7. RTS stands for Request to Send.

8. CTS stands for Clear to Send.
9. RTS and CTS can help reduce the hidden terminal problem.
10. CSMA CA cannot completely eliminate collisions.